

Who am I?

Breaking things since I was 12

Consulting Security Architect

I get to break things and help fix them

Other interests: traveling, hockey, snobby
beer, and my bulldog

#cyberspice

Let's talk dynamic security...

- Real-time threat intelligence
- Real-time threat analysis and detection
- Real-time threat response and remediation
- Real-time threat intelligence and detection
- Real-time threat response and remediation

Proactive responses based on patterns and threat intelligence

• Real-time threat intelligence- Real-time threat analysis and detection
- Real-time threat response and remediation
- Real-time threat intelligence and detection
- Real-time threat response and remediation

Networks are Failing Us

- | They work too well | They don't work
and enough |
|----------------------|-------------------------------|
| • They work too well | • They don't work |
| • They work too well | • They don't work |
| • They work too well | • They don't work |
| • They work too well | • They don't work |
- And I've seen that all over the world in the cloud

Software-Defined Networking

Software-defined networking (SDN) is a new approach to network architecture. It separates the control plane from the data plane, allowing for more flexible and programmable networks.

What's out there and how does it work?

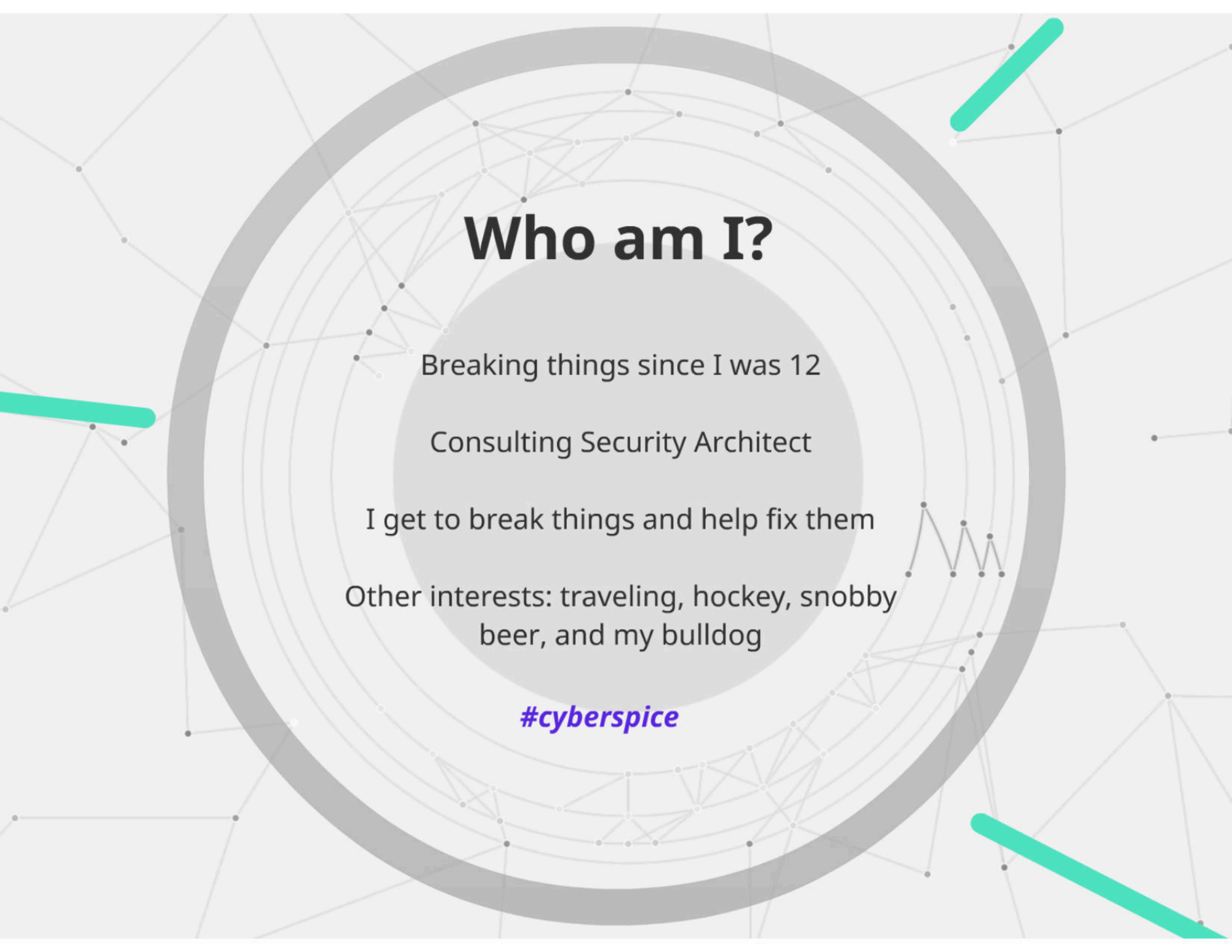
So why do we care?

Because we care about the security of our networks. We want to make sure that our networks are secure and that we can control them. We want to make sure that we can protect our data and our systems from attackers.

Let's break it down

Jon Medina
protiviti[®]
Take the Future with Confidence

Where are all the SDN Talks?
Because Skynet can defend itself now!



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Networks are Failing Us

They work too well

- Things are talking that shouldn't be
- We have to actively PREVENT systems from talking to each other
- Have you seen a firewall rule list lately?

They don't work well enough

- Applications have no network visibility
- Network hardware is getting complicated
- Infrastructure deployment is costly and inefficient
- Network devices are expensive

...And how does this all come together in the cloud?

Software Defined Networking

- Abstraction of the control plane and data plane from devices.
- The brains of network device functions are moved to a centralized location.
- This is *not* NFV



Switch Controller

ctions are

What's out there?

- Companies with closed source implementations:
 - Cisco Application Centric Infrastructure (ACI)
 - VMware NSX
- Companies with Open Source Open Flow implementations:
 - Juniper
 - HP
 - Arista
 - The Linux Foundation
 - ...and dozens more

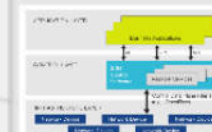
This is not a new technology
Google, Amazon, Facebook, NSA, most telcos have been using it for years

What's out there and how does it work?

Go with the flow



SDN Architecture



Core Concepts

- Controller sends flows (think ACLs) to switches
- Switches forward traffic based on flows
- If a switch doesn't have a flow that matches a packet, it asks the controller
- The controller either sends a new flow to the switch, or the packet is dropped



What's out there?

Companies with closed-source implementations:

- Cisco Application Centric Infrastructure (ACI)
- VMWare NSX

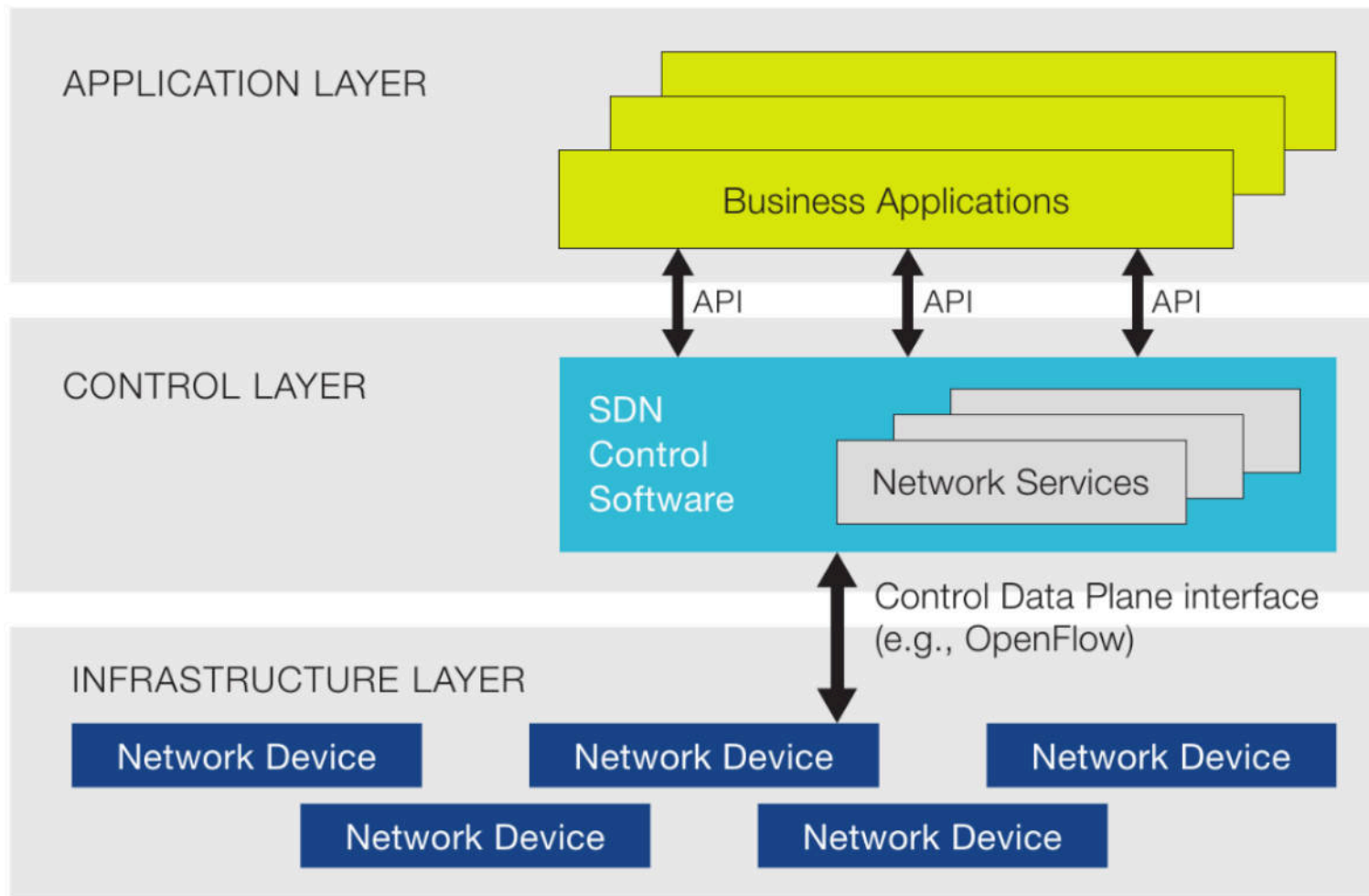
Companies with Open Source Openflow implementations:

- Juniper
- HP
- Avaya
- Big Switch
- ...and dozens more

This is not a new technology

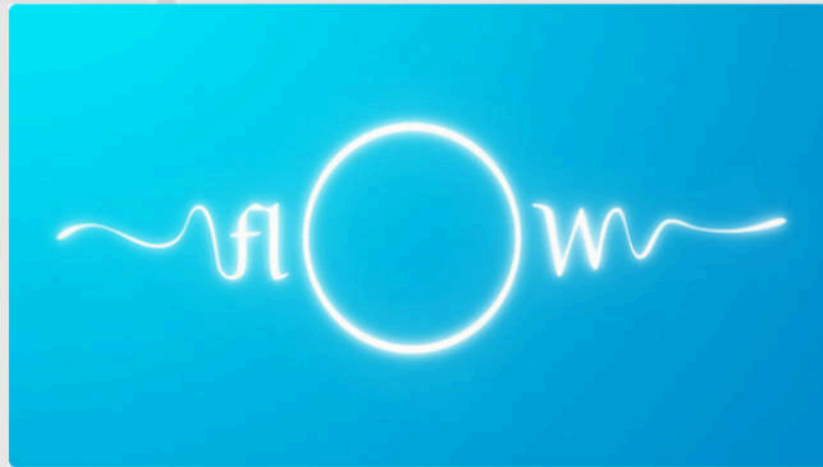
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SDN Architecture

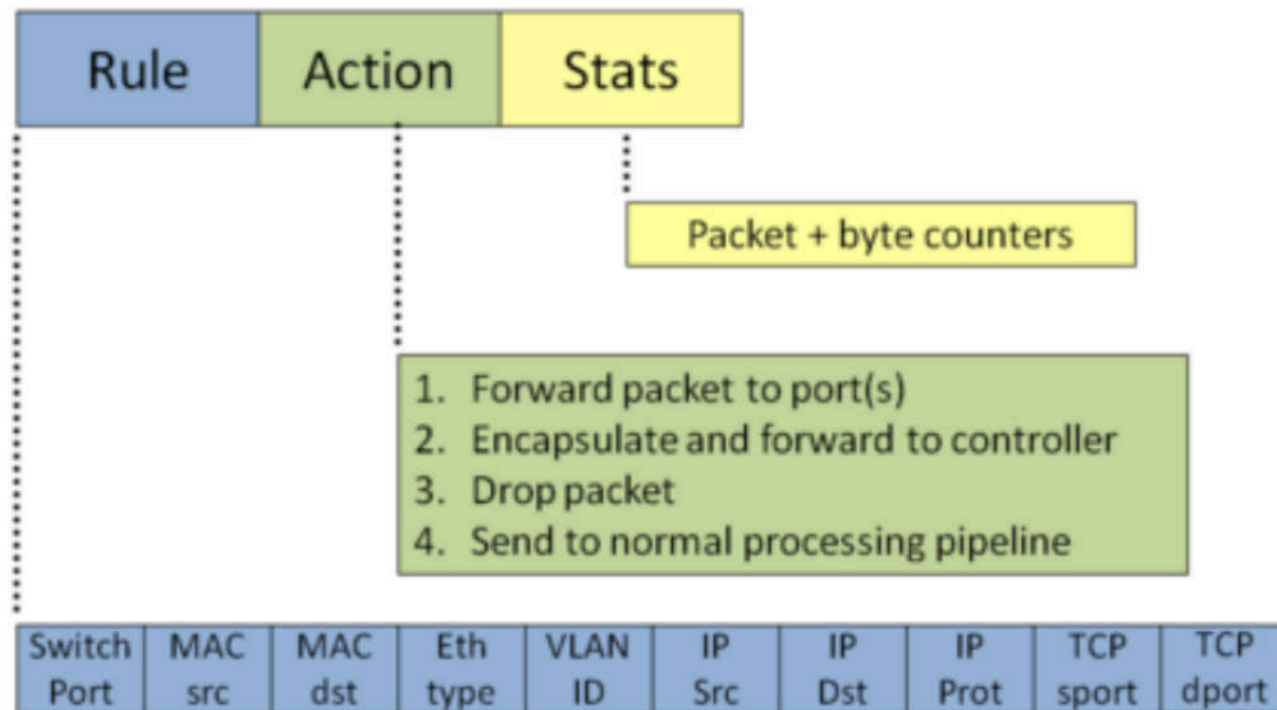


Core Concepts

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Go with the flow





Let's Check it Out

So why do we care?

- Absolute control over every aspect of network traffic
- Traffic policies can be applied globally to users and systems
- We can make dynamic decisions about how to treat network traffic based on every layer of the OSI model
- Flow tables created within flow tables

A background graphic featuring a network diagram with nodes and connecting lines, overlaid on a large, thick gray circle.

Security Use Cases

- What network ports are users ACTUALLY using? Do users really need to communicate with each other?
- How do you limit access by vendors and outsiders connected to your network?
- How do you control what devices are connected to your network, and how do you handle BYOD? IOT?

Let's talk dynamic security...

- Location based access control
- Multi-factor network authentication
- Granular control over application communications and cloud interactions.
- Vendor agnostic deployment assists in security tool-mageddon

Provision resources based on patterns using machine learning

- User and Service Account Behaviors
- Network traffic patterns
- Bandwidth consumption
- Application usage

- **Introducing....
Apache Spot (Incubating)**



What is Apache Spot?

- Spot is a community-driven project providing advanced analytics and comprehensive visibility across all security data using an open, scalable platform.
- Spot expedites threat detection, investigation and remediation via machine learning and consolidation of IT Telemetry into a comprehensive Data Lake.



<https://spot.incubator.apache.org/>

What does Spot Provide?

- Network Flows (nfcapd)
- DNS (PCAP)
- Proxy



TELEMETRY

- Open source decoders
- Load data in Hadoop



PARALLEL INGEST FRAMEWORK



MACHINE LEARNING

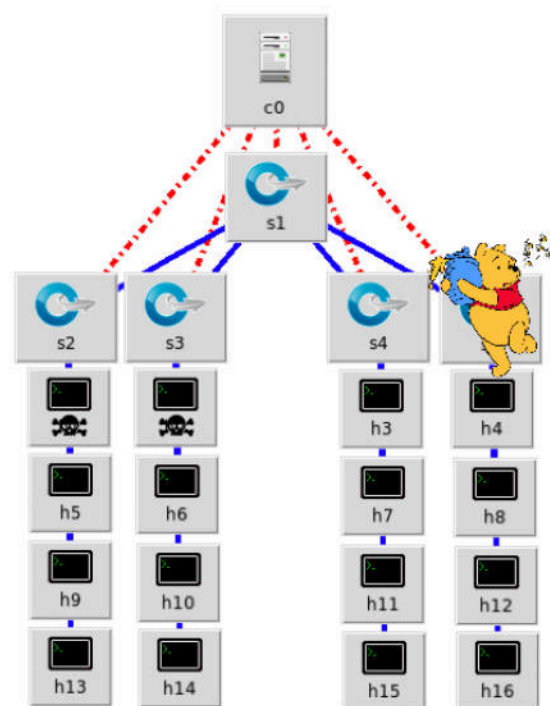
- Filter billions of events to a few thousand
- Unsupervised learning

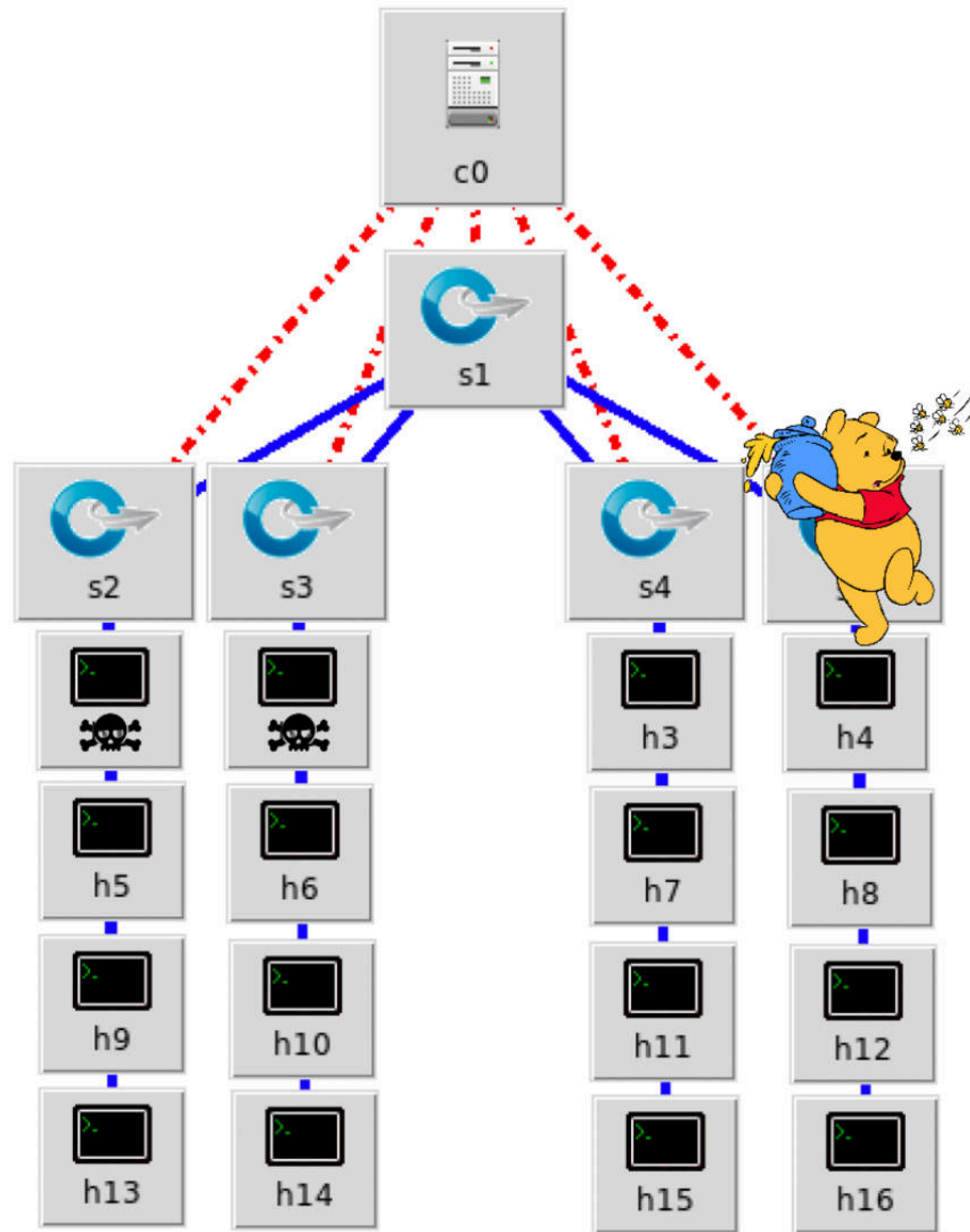


OPERATIONAL ANALYTICS

- Visualization, attack heuristics noise filter

Demo Architecture





Finally..... a helpful SDN tool

(hopefully)

- Takes sFlow information from a network
- Writes it to a mininet topology
- Communicates active flows to SDN controller
- Can be used to model or migrate production or lab environments

....**FloWrita**

github: @acksec

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Networks are Falling Us

They work how best
They don't work
well enough
...And how does this all come together in the cloud?

Software Defined Networking

What is it?
How does it work?
Why is it important?
What are the challenges?
What are the opportunities?

What's out there and how does it work?

So why do we care?

Because...
Because...
Because...
Because...
Because...

Let's talk dynamic security...

Traditional security models
are based on static configurations
and are not designed to adapt to
changing threats and attack vectors.

Proactive security based on
dynamic security building
- Real-time threat detection
- Automated response
- Continuous monitoring
- Adaptive security

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